

Fourier analysis – 4

Discrete Fourier analysis and characters

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Theoretical minimum.

Definition. A character of a finite abelian group G is a homomorphism $\chi: G \rightarrow \mathbb{C}^*$; for $G = \mathbb{Z}_N$ they are exactly $\chi_r(x) = e^{2\pi i r x / N}$. The characters form a group $\widehat{G}$ isomorphic to G , they are orthogonal: $\frac{1}{|G|} \sum_x \chi(x) \overline{\psi(x)} = \delta_{\chi\psi}$, and hence form an orthonormal basis of the space of functions on G .

Theorem 1 (the discrete transform). For $f: \mathbb{Z}_N \rightarrow \mathbb{C}$ put $\widehat{f}(r) = \sum_x f(x) e^{-2\pi i r x / N}$. Then $f(x) = \frac{1}{N} \sum_r \widehat{f}(r) e^{2\pi i r x / N}$, Parseval reads $\sum_x |f(x)|^2 = \frac{1}{N} \sum_r |\widehat{f}(r)|^2$, and the cyclic convolution satisfies $\widehat{f * g} = \widehat{f} \cdot \widehat{g}$. Counting solutions of an equation in $\mathbb{Z}_N$ is exactly evaluating a convolution, so it is exactly a computation with $\widehat{f}$.

Theorem 2 (the root-of-unity filter). $\frac{1}{n} \sum_{j=0}^{n-1} \omega^{jk}$ equals 1 when $n \mid k$ and 0 otherwise, so $\frac{1}{n} \sum_j F(\omega^j)$ extracts from a generating function the terms whose exponent is divisible by n . This is the case $G = \mathbb{Z}_n$ of Theorem 1 and was used for counting in Trig-2.

Fact 1 (Gauss sums). For an odd prime p the sum $G = \sum_{k=0}^{p-1} e^{2\pi i k^2 / p}$ has $|G| = \sqrt{p}$, and $G = \sum_k \left(\frac{k}{p}\right) e^{2\pi i k / p}$ is the Fourier transform of the Legendre symbol. Computed in Trig-2; here it is the model example of a character sum.

Fact 2 (large coefficients mean structure). $|\widehat{f}(r)|$ is large exactly when f correlates with the character χ_r ; if all non-trivial coefficients are small, the function is close to constant, which is the discrete counterpart of the Riemann–Lebesgue lemma and the basis of every “almost implies exactly” statement in this subject.

1. Prove the orthogonality relations for the characters of a finite abelian group, deduce the inversion formula and Parseval's identity for $\mathbb{Z}_N$, and prove that $\widehat{f * g} = \widehat{f}\widehat{g}$ for the cyclic convolution.
2. Using the discrete transform, count the number of subsets of $\{1, 2, \dots, n\}$ whose sum is divisible by m , and the number of solutions of $x_1 + \dots + x_k \equiv a \pmod{N}$ with all x_i in a prescribed subset $A \subseteq \mathbb{Z}_N$; express the answer through $\widehat{\mathbf{1}_A}$.
3. (OMOUS 2023) Let $f: \mathbb{Z}_N \rightarrow S^1$ satisfy $|f(x+y) - f(x)f(y)| < \varepsilon$ for all $x, y \in \mathbb{Z}_N$, where S^1 is the unit circle and ε is small. Prove that there is a character $g: \mathbb{Z}_N \rightarrow S^1$, that is $g(x+y) = g(x)g(y)$, with $|g(x) - f(x)| < \varepsilon$ for all x .
4. Let $A, B \subseteq \mathbb{Z}_N$. Prove that every element of $\mathbb{Z}_N$ is uniquely representable as $a+b$ with $a \in A, b \in B$ if and only if $|A| \cdot |B| = N$ and $\widehat{\mathbf{1}_A}(r)\widehat{\mathbf{1}_B}(r) = 0$ for every $r \neq 0$; deduce that in a tiling of $\mathbb{Z}_N$ the zero sets of the two transforms partition the non-zero frequencies. (The analogous statement for tilings of $\mathbb{Z}^2$ is problem 9 in the homework of Convex – 10.)
5. Let $A \subseteq \mathbb{Z}_p$ with $|A| = \alpha p$ and suppose $|\widehat{\mathbf{1}_A}(r)| \leq \delta p$ for every $r \neq 0$. Prove that the number of triples $(x, y, z) \in A^3$ with $x + z = 2y$ is $\alpha^3 p^2 + O(\delta p^2)$, and deduce that a set with small non-trivial Fourier coefficients contains a three-term arithmetic progression.